

3-Tier Churn Risk Intervention Playbook

RFM Analysis + Random Forest Classifier • 6,908 Customers Scored • 99.9% Model Accuracy

MODEL OUTPUT SUMMARY

6,908	3,550	3,356	2	99.9%
Customers Scored	High Risk (51.4%)	Medium Risk (48.6%)	Low Risk (0.03%)	Model Accuracy

SCORING METHODOLOGY

Customers were scored using the industry-standard RFM (Recency, Frequency, Monetary) framework, then classified into three churn risk tiers using a Random Forest machine learning model trained on 128,975 Amazon India transactions. Each customer segment receives a composite RFM score (3-9 scale) and is assigned to a risk tier with a specific, time-bound intervention action.

3-TIER INTERVENTION FRAMEWORK

■ HIGH RISK

3,550 customers | 51.4% | RFM Score ≤ 4

→ ACTION: Win-back email + 15% discount voucher within 48hrs

Signal	No order in 40+ days, low frequency, below-avg spend
Urgency	IMMEDIATE — contact within 48 hours of identification
Channel	Personalised email + SMS + push notification
Offer	15% discount voucher on their most purchased category
Goal	Re-engage before 60-day churn threshold is crossed
Success KPI	Re-purchase within 30 days of intervention
Owner	CRM Team + Performance Marketing

■ MEDIUM RISK

3,356 customers | 48.6% | RFM Score 5-6

→ ACTION: Personalised product recommendation push notification

Signal	No order in 20-40 days, moderate frequency
Urgency	Within 7 days of risk identification
Channel	Push notification + in-app banner + email
Offer	Curated product recommendations based on past purchases
Goal	Maintain engagement before they drop to High Risk tier
Success KPI	Click-through on recommendation within 14 days
Owner	Product Team + Email Marketing

■ LOW RISK

2 customers | 0.03% | RFM Score ≥ 7

→ **ACTION: Loyalty points reward + early access to new arrivals**

Signal	Recent order, high frequency, above-avg spend
Urgency	Monthly loyalty touchpoint — no urgency
Channel	Email newsletter + loyalty programme dashboard
Offer	Double loyalty points on next purchase + early access
Goal	Deepen engagement and increase order frequency
Success KPI	Increase in monthly order frequency by 10%
Owner	Loyalty Programme Team

IMPLEMENTATION TIMELINE

WEEK 1	WEEK 2-3	WEEK 4	MONTH 2-3
- Export High Risk list - Set up win-back email - Brief CRM team	- Launch Medium Risk push - Monitor High Risk re-purchase - A/B test discount %	- Review campaign results - Adjust messaging - Update risk scores	- Scale successful campaigns - Re-score all customers - Report to leadership

Juvana Dsouza | B.E. AI & Data Science, Fr.
CRCE Mumbai

github.com/juvana81/ecommerce-churn-risk-scorer

Tools: Python • Scikit-learn • RFM •
Power BI